Jordan M. Zabrecky

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EDUCATION

Ph.D., Natural Resources and Environmental Science 2022-present
University of Nevada, Reno
Advisor: Dr. Joanna Blaszczak

B.S., Geological Sciences *with highest distinction* 2016-2020
University of North Carolina at Chapel Hill
Minors in Environmental Science & Studies and Marine Science

RESEARCH & PROFESSIONAL EXPERIENCE

2022-	**NSF Graduate Research Fellow** Department of Natural Resources and Environmental Sciences, University of Nevada, Reno
2021	**DoE Student Undergraduate Laboratory Intern** Subsurface Processes Group, Pacific Northwest National Lab
2018-2020	**Undergraduate Research Assistant** Department of Geological Sciences, University of North Carolina at Chapel Hill
2018	**NSF IDEA Summer Undergraduate Researcher** UNC Institute for the Environment & Department of Geological Sciences, University of North Carolina at Chapel Hill

FELLOWSHIPS & AWARDS

2023-2026	**Graduate Student Association Travel Awards** *(x4)*, University of Nevada, Reno	$1,800 *(total)*
2022-2027	**Graduate Research Fellowship**, National Science Foundation	$147,000
2022-2023	**Graduate Dean's Fellowship**, University of Nevada, Reno *(1 of 6 awarded)*	$40,000
2019	**Charles & Elaine Mims Fellowship**, Department of Geological Sciences, University of North Carolina at Chapel Hill	$3,455
2016-2019	**Dean's List**, University of North Carolina at Chapel Hill *(awarded every full-time semester)*	
2016-2020	**Special Scholarship**, National Merit	$4,000

RESEARCH SUPPORT

2026 *University of Nevada, Reno Graduate Student Association* $1,000
Research, Travel, and Materials Grant
Influence of habitat when forecasting cyanobacterial harmful algal blooms (cHABs) within a complex delta

2023 *Society for Freshwater Science* $1,000
General Endowment Award

Environmental controls on the growth and toxin production of benthic cyanobacteria *Microcoleus* and *Anabaena* in rivers of northern California, USA

2023 *Nevada IDeA Network of Biomedical Research Excellence (INBRE)* $4,000
Nevada Bioinformatics Center Service Award
Furthering our understanding of neurotoxin-producing benthic riverine cyanobacteria that pose a public health risk: Investigating species co-existence

PUBLICATIONS

Peer-Reviewed Publications:

3. Kelly, L.T., S.A. Wood, [and 32 others including **J.M**. **Zabrecky**]. 2026. Tracking the global proliferation of benthic aquatic *Microcoleus*: taxonomy, distribution, toxin production, ecology, and future directions. *Water Research* 249, 125441.
   DOI: 10.1016/j.watres.2026.125441
2. Brown, S.M., J.R. Blaszczak, R.K. Shriver, R.C. Jones, A. Sohrab, R. Goel, G.L. Boyer, B. Wei, K.M. Manoylov, T.R. Nelson, **J.M. Zabrecky**, and R. Stancheva. 2025. Growth and anatoxin-a production of *Microcoleus* (Cyanobacteria) strains from streams in California, USA. *Harmful Algae* 144, 102834.
   DOI: 10.1016/j.hal.2025.102834
1. **Zabrecky, J.M.**, X-M. Liu, Q Wu, and C. Cao. 2021. Evidence of anthropogenic gadolinium in Triangle Area Waters, North Carolina, USA. *Water* 13(14), 1895.
   DOI: 10.3390/w13141895

Data Releases:

1. **Zabrecky, J.M.**, T.A. Elliott, M. Hickey, H. Lei, R.S. Christova, G. Boyer, L. Genzoli, G. Johnson, and J.R. Blaszczak. 2025. Anatoxin concentrations, algal assemblages, and water quality data for the South Fork Eel, Salmon, and Russian Rivers in northern California, 2022-2023. *Environmental Data Initiative*.
   DOI: 10.6073/pasta/5175ff92803eeaaac31d5fd724108d92

Other Publications

1. **Zabrecky, J.M.** and T.A. Elliott. 2026. Salmon River Coloring Book. *Open Educational Resources Commons* 141536.
   oercommons.org/courseware/lesson/141536

Manuscripts *In Prep, In Review, or In Press*

- **Zabrecky, J.M.**, T.A. Elliott, M. Hickey, K. Bouma-Gregson, G. Boyer, R. Fadness, L. Genzoli, R. Goel, G. Johnson, R. Shriver, R. Stancheva, M. Thomas, Z. Triumph, and J.R. Blaszczak. Predicting taxon-specific benthic cyanobacterial mat cover and anatoxin concentrations in northern California rivers. *In Review*.
  preprint DOI: 10.64898/2026.09.04.748937
- **Zabrecky, J.M.**, T.A. Elliott, M. Hickey, K. Bouma-Gregson, G. Boyer, R.M. Fitzpatrick, R. Goel., G. Johnson, F. Manzoor, A. Sohrab, R. Stancheva, Z. Triumph, and J.R. Blaszczak. Algal and bacterial assemblages associated with benthic cyanobacteria *Microcoleus* and *Anabaena* vary among rivers and with anatoxin detection. *In Review.*

PRESENTATIONS

Conference Presentations

Zabrecky, J.M., T. Elliott, M. Hickey, R. Fitzpatrick, R. Stancheva, L. Genzoli, S. Kaiser, A. Sohrab, R. Goel, and J.R. Blaszczak. Spatiotemporal variation in algal and bacterial assemblages associated with toxigenic benthic cyanobacteria *Microcoleus* and *Anabaena* in northern Californian rivers. (Poster). *13th U.S. Symposium on Harmful Algae*. Cedar Rapids, IA. October 2026.

Zabrecky, J.M., T. Elliott, M. Hickey, R. Fitzpatrick, R. Stancheva, L. Genzoli, S. Kaiser, A. Sohrab, R. Goel, and J.R. Blaszczak. Spatiotemporal variation in algal and bacterial assemblages associated with toxigenic benthic cyanobacteria *Microcoleus* and *Anabaena* in northern Californian rivers. (Oral). *Society for Freshwater Science Annual Meeting*. Spokane, WA. May 2025.

Zabrecky, J.M., T. Elliott, M. Hickey, R. Stancheva, G. Boyer, K. Bouma-Gregson, L. Genzoli, R. Fadness, M. Thomas, R. Goel, R. Shriver, and J.R. Blaszczak. Predicting taxa-specific benthic cyanobacteria occurrence and anatoxin concentrations in a recreational river. (Poster). *Ecological Forecasting Initiative 2025 Conference*. Blacksburg, VA. May 2025.

Zabrecky, J.M., T. Elliott, M. Hickey, R. Stancheva-Christova, K. Bouma-Gregson, L. Genzoli, R. Fadness, M. Thomas, G. Boyer, S. Kaiser, A. Sohrab, R. Goel, R. Shriver, and Blaszczak, J.R. Assessing the synchronicity of anatoxin-producing benthic cyanobacteria and river ecosystem productivity. (Poster). *12th U.S. Symposium on Harmful Algae*. Portland, ME. October 2024.

Zabrecky, J.M., T. Elliott, M. Hickey, R. Stancheva-Christova, K. Bouma-Gregson, L. Genzoli, R. Fadness, M. Thomas, G. Boyer, S. Kaiser, A. Sohrab, R. Goel, R. Shriver, and J.R. Blaszczak. Assessing the synchronicity of anatoxin-producing benthic cyanobacteria and river ecosystem productivity. (Oral). *Society for Freshwater Science Annual Meeting*. June 2024.

Zabrecky, J.M., M. Hickey, R. Stancheva, K. Bouma-Gregon, L. Genzoli, G. Johnson, R. Fadness, S. Kaiser, A. Sohrab, R. Goel, R. Shriver, and J.R. Blaszczak. Environmental controls on the growth and toxin production of benthic cyanobacteria Microcoleus and Anabaena in rivers of northern California, USA. (Poster). *Society for Freshwater Science Annual Meeting*. Brisbane, AU. June 2023.

Zabrecky, J.M., H.P. Emerson, and C.I. Pearce. Geochemical controls on historic Pu releases in a radioactive waste disposal trench. (Oral). *Pacific Northwest National Lab July Symposium*. Virtual. July 2021.

Zabrecky, J.M. and X-M. Liu. Anthropogenic gadolinium in Triangle Area waters, NC: a case study of wastewater treatment plants. (Poster). *Geological Society of America National Conference*. Indianapolis, IN. November 2018.

Community & Webinar Talks *(*denotes co-presenters)*

Davidson, J.* and **J.M. Zabrecky***. Predicting benthic cyanobacterial proliferations and anatoxin concentrations in a northern Californian river. *California cHABs Quarterly Meeting*. Virtual. August 2026.

Blaszczak, J.* and **J.M. Zabrecky***. Macroinvertebrate communities within benthic anatoxin-producing cyanobacterial mats. Environmental Protection Agency Benthic HABs Discussion Group. Virtual. March 2025.

Zabrecky, J.M. Predicting the timing of toxin-producing benthic cyanobacteria in rivers. *Ecological Forecasting Initiative Student & Early Career Society: Introduction to Ecological Forecasting Graduate Student Flash Talks and Q&A*. Virtual. February 2025.

Zabrecky, J.M. Benthic cyanobacteria in rivers of northern California. Headwaters Science Institute and Serene Lakes Community. Soda Springs, CA. July 2024.

TEACHING & MENTORING

Academic Teaching Experience

Spring 2026 **Instructor of Record**, EECB 790: Code Review Seminar, University of Nevada, Reno
- Co-developed and co-led seminar with another graduate student, Ellen McMullen, to offer students university credit to participate in peer code review
- Seminar included peer code review with interspersed readings, workshops, and lectures related to the code review process in ecology and beyond

Winter 2026 **Teaching Assistant**, Intro to R Workshop, University of Nevada, Reno

Teacher & Mentor Training

Fall 2025 ITLD 720: Instructional Design Theory and Application, University of Nevada Reno

Youth Environmental & Outdoor Education

Summer 2024 **Young Scientist Exploration Camp Lead Mentor**, Headwaters Science Institute, Soda Springs, CA
- Led grade 3 students through a week-long science project and various supplemental science activities related to plants and soil

Spring 2022 **Outdoor Nature Educator**, Hitchcock Center for the Environment, Amherst, MA
- Designed and led week-long February & April vacation camp activities including environmental education activities, games, and outdoor play for grades 3-5

Spring 2022 **Nature Play Afterschool Program Volunteer**, Hitchcock Center for the Environment, Amherst, MA
- Supervised and engaged in outdoor play for grades 1-3

Rock Climbing Instruction & Coaching

2021-2022 **Youth Competition Team Coach**, Central Rock Gym, Hadley, MA
- Assisted leading team practices and coaching at competitions for a nationally competitive youth climbing team

2018-2020 **Youth Elite Team Coach**, Progression Climbing, Chapel Hill, NC

2017-2018 **Youth Intermediate Team Coach**, Progression Climbing, Chapel Hill, NC
- Developed and led practices to strengthen youth climbers' physical and technical skills to progress to the nationally competitive "elite" team

2017-2019 **Team Captain**, UNC Climbing Team, Chapel Hill, NC
- Developed and led team practices, organized team competitions schedule, planned travel itineraries, and managed related team documents
- Led team to regional championship titles in both Speed and Sport Climbing in 2018 and 2019 for a total of four titles
- Placed top sixteen at collegiate national three times and won two regional champion titles in Bouldering and Speed Climbing as an individual competitor

Spring 2017 **Climbing Instructor**, UNC Campus Recreation, Chapel Hill, NC

OUTREACH, SERVICE, & PROFESSIONAL MEMBERSHIPS

Service

2026- **Ecological Forecasting Initiative (EFI) Steering Committee**, EFISECA Representative

2025- **Ecological Forecasting Student & Early Career Association (EFISECA)**, Cochair

2024-2025 **EECB/NRES CodeClub,** Co-Leader
- This graduate student club was subsequently developed into a seminar course (see "Teaching & Mentoring") to offer academic credit for participation

2023-2024 **California Benthic HABs Guidance Subcommittee**, Member

Outreach Activities

2025-2026 **Letters to a Pre-Scientist Program**, Pen Pal Program
Summer 2024 **NevadaFIT (Freshman Intensive Transition Program) Water Quality Activity**, University of Nevada, Reno
Spring 2023 **Nevada STEM Day**, University of Nevada, Reno
Spring 2019 **FEMMES Day Camp**, University of North Carolina at Chapel Hill
Spring 2019 **STEMville Family Science Nights**, Franklin County Schools, NC

Professional Memberships

Ecological Forecasting Initiative (EFI), Society for Freshwater Science (SFS)

Nonprofit Fundraising

2022 **Starbucks Foundation's Neighborhood Grant** *to* the Hitchcock Center for the Environment, Amherst, MA $1,000

Reviewer

San Francisco Estuary and Watershed Science

Last updated November 2026